

An improved machining temperature prediction model for aerospace alloys: Effect of cutting edge radius and tool wear

Jonathan Theraroz^a, Oguzhan Tuysuz^{a,*}, Julius Schoop^b

^a Department of Mechanical Engineering, Polytechnique Montréal, 2500, Chemin de Polytechnique, H3T 1J4 Montréal, QC, Canada

^b Institute for Sustainable Manufacturing, Department of Mechanical & Aerospace Engineering, University of Kentucky, Lexington, KY 40506, USA

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ABSTRACT

Temperature rise during machining impacts the workpiece material properties, residual stresses, surface and sub-surface quality. Experimental, numerical, and analytical methods have been used to predict the temperature fields in the tool, workpiece and chip. Each approach has its limitations: experimental techniques are cumbersome with expensive equipment, and numerical modeling is computationally inefficient. Existing analytical models only consider the effect of wear while ignoring the edge radius, though the latter changes with the flank wear in practice. To address this limitation, this article proposes an improved analytical temperature prediction model for orthogonal machining by introducing discrete linear heat sources on the edge radius of the cutting edge. The model describes the machining deformation zones by moving or stationary heat sources and models the adiabatic surfaces by imaginary heat sources. The heat partition is calculated to describe the amount of temperature transferred from a heat source to a given body. A global coordinate system is introduced to facilitate the integration of the edge radius in the temperature model, and variation in the direction of the heat source velocity. Temperature predictions of the developed model were experimentally verified using an inverse method based on XRD residual stress measurements. The results of the analysis show that the proposed model is reasonably accurate and most importantly computationally efficient alternative to tedious experimental measurements or more complicated finite element approaches.

1. Introduction

There has been a great deal of work on prediction and measurement of machining temperature in the last two decades due to its effect on the surface quality, tool life and built-up edge formation [1,2]. Several experimental methods [3] using thermal cameras [4] or thermocouples [5,6] have been developed to measure the temperature in the tool and in the workpiece. However, detailed temperature measurements require complex and expensive set-ups [1,2]. Numerical modeling using finite elements (FE) and finite difference (FD) provides accurate temperature predictions [7–9] and is therefore the most widespread simulation method.

Halfway between the experimental and numerical methods, analytical models were first developed by Jaeger, Rosenthal, Chao and Trigger [10–12], and have been the basis of most of the subsequent analytical models. Komanduri and Hou [13–15] refined these models and predicted the temperature rise in orthogonal cutting in the primary (shear) and secondary (chip-rake face interface) deformation zones. Fig. 1

shows three deformation zones occurring during machining.

A significant amount of work has been conducted on calculation of the heat partition coefficients of the heat sources, which describes the amount of energy going into each respective body at the interface. Initially, a constant heat partition was applied at the shear [16] and chip-rake [17] interfaces, but it was later adjusted to be non-uniform [13]. Two methods have been proposed in literature to calculate the non-uniform heat partition. The first method [14] formulates the heat partition and optimizes its constants ($B_{0,q1}$, $B_{0,q2}$, ΔB , C , m , k) to match the temperatures on both sides of the interface:

$$B_{q1}(l_i) = B_{0,q1} - \Delta B + 2\Delta B \left(\frac{l_i}{L}\right)^m + C\Delta B \left(\frac{l_i}{L}\right)^k \quad (1)$$

$$B_{q2}(l_i) = B_{0,q2} + \Delta B - 2\Delta B \left(\frac{l_i}{L}\right)^m - C\Delta B \left(\frac{l_i}{L}\right)^k \quad (2)$$

where q_1 and q_2 are the two sides of the interface (e.g., q_1 = chip and

* Corresponding author.

E-mail address: oguzhan.tuysuz@polymtl.ca (O. Tuysuz).

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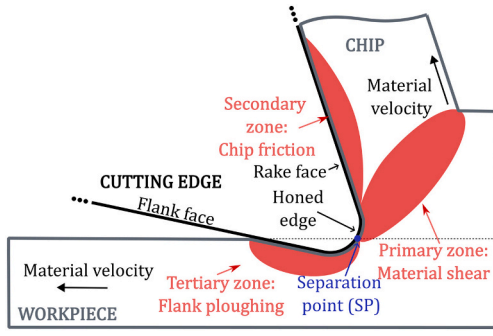


Fig. 1. Deformation zones in machining.

$q_2 = \text{tool}$), $B_{q_1}(l_i)$ and $B_{q_2}(l_i)$ are the position-dependant heat partitions of the heat sources on each side, l_i is the position on the heat sources of length L . A second method proposed by Huang and Liang [18] discretizes the heat sources into infinitesimal elements with constant heat partitions. It then solves the linear equations of temperature equality at all differential elements to determine the non-uniform heat partitions. Some improvements to [13–15] have been lately proposed. Chou and Song [19] added the contribution of the tertiary deformation zone (flank rubbing) to the heat generation. This was later repeated by Yan et al. [20] using an empirical heat partition from an infrared thermal camera in that zone. Huang and Yang [21] introduced the heating time to take into account the workpiece's cooling cycle. Feng et al. [22] included the loss of contact between the workpiece and the cutting tool during ultrasonic vibration-assisted milling. The standard model of [13–15] was improved by incorporating the microstructure evolution influenced by the material temperature and strain into the heat intensity calculation [23], as well as by adding the laser preheating to the standard machining temperature prediction model [24].

Though there exist analytical temperature prediction models, the effect of edge radius, which is a crucial component of the tool wear in this work, on the temperature rise during machining has not been addressed. Therefore, this study proposes a method to integrate the effect of edge radius and varying heat source velocity direction into the steady state temperature prediction model originally proposed by Komanduri and Hou [13–15] for orthogonal machining. The main objective of the current paper is to predict the workpiece temperature at the interface between the cutting edge and the work material, which gives the peak workpiece temperature required in chatter stability calculations using the analytical process damping models [25,26].

2. Existing temperature rise model and its limitations

The original form of the steady-state temperature prediction model [19–21] described in this section assumes that: the cutting is orthogonal (two-dimensional-2D); free surfaces are adiabatic; the tool is perfectly sharp, which is lifted in this article; work material is homogenous; plastic deformations and frictional forces are entirely converted into heat without any loss.

2.1. Definition of the heat source

Three deformation zones are considered: the shear plane, the rake-chip friction, and the flank ploughing [13,18,20]. Three distinct bodies are considered: the chip, the workpiece, and the tool. Each deformation zone is an interface between two bodies (including the shear plane as the workpiece/chip interface) and is modeled by two heat sources (one in each body) of specific heat intensity and length as exemplified for chip-rake face interface in Fig. 2. Two types of heat sources are used depending on the body it describes: a moving line for the moving material on the workpiece and chip side, and a stationary rectangle on the tool side [15] as illustrated in Fig. 2.

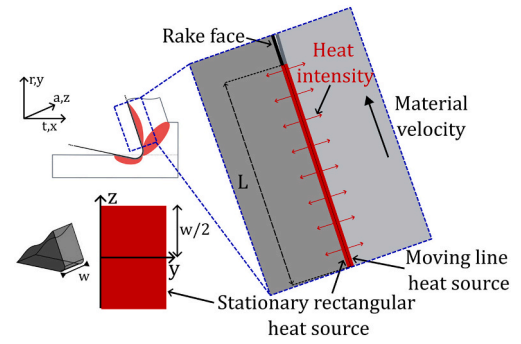


Fig. 2. Example of heat source at chip-rake interface.

2.2. Temperature rise calculation

The temperature distribution around the heat sources is governed by Fourier's heat transfer equation [11,27,28]. That provides two relations (for stationary rectangular and moving line heat sources) describing the temperature rise $\theta = T - T_{init}$ from an initial temperature T_{init} at a point of interest $M(x, y, z)$ with one imaginary heat source, i.e. one adiabatic surface, defined symmetric to the original one with the adiabatic surface as symmetry plane (method of images):

$$\theta(x, y) = \int_0^L \frac{q(l_i)}{2\pi\kappa} e^{-R_v V_h / 2a} \left[K_0 \left(\frac{V_h R}{2a} \right) + K_0 \left(\frac{V_h R'}{2a} \right) \right] dl_i \quad (3)$$

for a moving line heat source, and,

$$\theta(x, y, z) = \int_0^L \frac{q(l_i)}{2\pi\kappa} \int_{-w/2}^{w/2} \left(\frac{1}{R} + \frac{1}{R'} \right) dz_i dl_i \quad (4)$$

for a rectangular heat source. $a = \kappa / \rho c_p$ is the thermal diffusivity of the material in which the heat source is located, κ is the thermal conductivity, ρ is the density, and c_p is the specific heat; L is the length of the heat source; w is the width of the rectangular heat source (width of cut in orthogonal machining); γ is the tilt angle; V_h is the material flow velocity at the interface (for the moving heat source); $q(l_i)$ is the heat intensity at l_i ; R and R' are the distances between the point of interest $M(x, y, z)$ and the position at l_i in the original and imaginary heat sources, respectively; R_v is the projection of R in the direction of the heat source motion; and $K_0(x)$ is the Bessel function of second kind of order zero.

A general definition of heat intensity is given as [28–30]:

$$q = \frac{F V_h}{w L} \quad (5)$$

with F is the force occurring at the interface, V_h is the material flow velocity at the interface, and L is the length of the interface, i.e. length of the heat source. The heat intensity is usually defined as constant for the shear and flank interfaces, but has been modified as non-uniform to represent the stick and slide behavior of the chip on the rake face of the tool [28,31]:

$$q_{f,r}(l_i) = \begin{cases} q_r \frac{2}{1+r_s}, & 0 < l_i < r_s L_c \\ q_r \frac{2(L_c - l_i)}{(1-r_s^2)L_c}, & r_s L_c < l_i < L_c \end{cases} \quad (6)$$

where q_r is the heat intensity of the rake-chip interface defined by Eq.(5), l_i is the position on the heat source, L_c is the chip-rake contact length and also length of the heat source, r_s defines the stick and slide portion of the interface and is taken as zero [28] in this work (see Fig. 3).

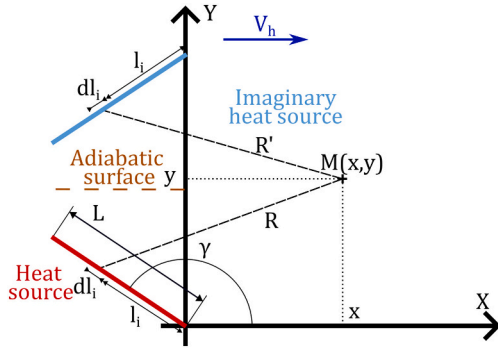


Fig. 3. Visual representation of geometric parameters of temperature calculation using a moving line heat source.

2.3. Heat partition calculation

When two heat sources are considered at both sides of an interface, the temperature at that interface can be discontinuous due to the difference of heat source geometry and the contribution of the other heat sources. The discontinuity causes heat flow at the interface, which invalidates the assumption of steady-state temperature [28]. A heat partition is therefore necessary to match the temperature at all contacting interfaces to ensure steady-state conditions. While some constant heat partition values were first considered, non-uniform distribution was quickly adopted due to the complexity of the temperature field [13]. Integration of the non-uniform heat partition in Eqs. (3) and (4) yields:

$$\theta(x, y) = \int_0^L B(l_i) \frac{q(l_i)}{2\pi\kappa} e^{-R_y V_h / 2a} \left[K_0 \left(\frac{V_h R}{2a} \right) + K_0 \left(\frac{V_h R'}{2a} \right) \right] dl_i \quad (7)$$

$$\theta(x, y, z) = \int_0^L B(l_i) \frac{q(l_i)}{2\pi\kappa} \int_{-w/2}^{w/2} \left(\frac{1}{R} + \frac{1}{R'} \right) dz_i dl_i \quad (8)$$

where $B(l_i)$ is the heat partition of the heat source at l_i .

Non-uniform heat partition has been adopted by numerous researchers [18,28,29,31–34], and two main analytical methods are used to define it. The most common one was proposed by Huang and Liang [18], which discretizes the heat sources into infinitesimal elements, and used by several researchers [28,29,33,34]. An arbitrary heat partition distribution is chosen at all interfaces, so that their sum is equal to 1 (e. g., for the rake-chip interface at l_i , the heat partition will be $B_f(l_i)$ on the chip side and $1 - B_f(l_i)$ on the tool side). The non-uniform heat partition is then assumed to be constant within each infinitesimal element (B_f and $1 - B_f$). By ensuring the temperature equality at each infinitesimal element (considering the contributions from all the other heat sources contributing to the temperature of the element), a linear system of equations is obtained. Solving that system provides the heat partition

value for each infinitesimal element, thereby recreating the non-uniform heat distribution. This method is illustrated in Fig. 4 via the example of the heat source on the chip side of the rake-chip interface.

For example, the temperature from both the tool and chip sides must be equal for the rake-chip interface shown in Fig. 4. The heat sources are therefore discretized into infinitesimal elements with a constant heat partition, which leads to:

$$\theta'_{bm}(x_i, y_i) = B_m \int_{l_i}^{l_{j+1}} \frac{q(l_i)}{2\pi\kappa} e^{-R_y V_h / 2a} \left[K_0 \left(\frac{V_h R}{2a} \right) + K_0 \left(\frac{V_h R'}{2a} \right) \right] dl_i \quad (9)$$

for a moving line heat source, and

$$\theta'_{bm}(x_i, y_i, z_i) = B_m \int_{l_i}^{l_{j+1}} \frac{q(l_i)}{2\pi\kappa} \int_{-w/2}^{w/2} \left(\frac{1}{R} + \frac{1}{R'} \right) dz_i dl_i \quad (10)$$

for a stationary rectangular heat source, where $\theta'_{bm}(x_i, y_i)$ is the temperature rise at l_i on the rake-chip interface, on the side $b \in \{c, t\}$ from the infinitesimal heat source element dl_i at the interface $m \in \{p, f, s\}$ (c and t refer to the chip and the tool, respectively, and p, f and s refer to the ploughed, chip-rake friction, and shear interfaces); $[l_j, l_{j+1}]$ is the infinitesimal element of the heat source at the interface m . At one infinitesimal element on the rake-chip interface, equating the temperature from the heat sources at both side of the interface yields:

$$B_p \theta'_p(l_i) + B_s \theta'_s(l_i) + B_f (\theta'_{cf}(l_i) + \theta'_{tf}(l_i)) = \theta'_p(l_i) + \theta'_f(l_i) \quad (11)$$

A similar approach can be followed for all the other interfaces, which provides a set of linear equations that can be put in a matrix form when all the infinitesimal elements and all the interfaces are considered:

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}_{3n \times 3n}, B = \begin{bmatrix} B_p \\ B_f \\ B_s \end{bmatrix}_{3n \times 1}, C = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix}_{3n \times 1} \quad (12)$$

where $A_{IJ}(i, j)$ corresponds to the temperature rise at l_i in the interface I due to the heat source element $[l_j, l_{j+1}]$ in the heat source J and its corresponding equations. $C_I(i)$ corresponds to temperature rise at l_i in the interface I of all the heat sources being the constant terms of the linear equation. $B_I(i)$ corresponds to the heat partition at l_i in the interface I of two corresponding heat sources and n corresponds to the number of discretized heat source elements.

2.4. Limitations of the existing model

Though there are several studies in literature, the existing models have only considered sharp tools without an edge radius. However, estimation of the temperature under the edge (hone) radius region is necessary for this work due to the focus on flank wear and where the edge radius that reaches up to $55 \mu\text{m}$ for a flank wear of $250 \mu\text{m}$ in this work. Hence, it is imperative to develop a method to add the contribution of the edge radius in the temperature prediction as proposed in this study. Moreover, more general temperature prediction equations are introduced using a global coordinate system rather than a local one for each heat source, which allows variations in the direction of heat source velocity.

3. Improved temperature prediction model

3.1. Linear approximation of the honed edge

Although the circular edge radius impacts the ploughing forces, there are no studies included its effect in the analytical prediction of the workpiece temperature during machining. The approach to include the hone radius consists of approximating its circular geometry by a set of line segments to use Eqs. (7) and (8). The hone radius is divided into two

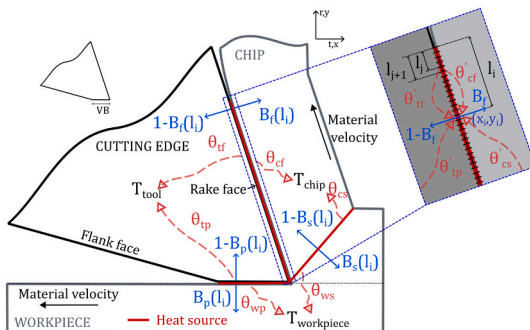


Fig. 4. Example of heat distribution on the rake-chip interface of a sharp but worn tool (flat wear length of VB).

main zones: the portion below the material separation point (SP) and the portion above it to differentiate the material velocity and its direction in both regions. These two regions are respectively discretized into n_d and n_u elements as illustrated in Fig. 5.

In order to use Eqs. (7) and (8), the length and angle of each tilted line are defined in the upper and lower regions. The i^{th} tilted line in the lower region is defined by $\gamma_{d,i}$ and $L_{d,i}$ as:

$$\gamma_{d,i} = \frac{3\pi}{2} - \beta_{d,i} - \frac{d\beta_d}{2} \quad (13)$$

$$L_{d,i} = \sqrt{\left(-R_e \sin(\beta_{d,i} + d\beta_d) + R_e \sin(\beta_{d,i})\right)^2 + \left(R_e \cos(\beta_{d,i} + d\beta_d) - R_e \cos(\beta_{d,i})\right)^2} \quad (14)$$

with

$$d\beta_d = \frac{\chi}{n_d} \quad (15)$$

$$\beta_{d,i} = \left(\frac{\pi}{2} - \chi\right) + d\beta_d(i-1) \quad (16)$$

$$E_{d,i} = (R_e \cos \beta_{d,i} + O_x, -R_e \sin \beta_{d,i} + O_y) \quad (17)$$

where χ is the separation angle, and R_e is the radius of the honed edge. The coordinates of the point O are defined as:

$$O(t) = (-R_e \sin \chi, R_e \cos \chi) \quad (18)$$

The i^{th} tilted line segment in the upper region is similarly defined by $\gamma_{u,i}$ and $L_{u,i}$. Eqs. (13), (14) and (17) can be reused, and Eqs. (15)–(16) are modified as:

$$d\beta_u = \frac{\pi/2 - \chi + \alpha_r}{n_u} \quad (19)$$

$$\beta_{u,i} = -\alpha_r + d\beta_u(i-1) \quad (20)$$

It must be noted that accurate approximation of the edge radius requires a sufficient number of heat sources (more than three heat sources are typically required, though this may need to be increased for larger

edge radii) on both regions.

3.2. Modification of the temperature rise equations

In literature, a local coordinate system is used for each heat source to calculate the temperature rise. This method is practical for a limited number of heat sources, but becomes rather complex when the heat sources are numerous and vary greatly in position and orientation as in this study, due to the linear approximation of the circular heat source around the edge radius. Therefore, a global coordinate system is proposed.

Existing studies use a constant velocity direction and amplitude for

any heat sources along an axis of the local coordinate system or in parallel to the symmetry axis. To address the challenge in modeling several heat sources in this work, the temperature rise equations are modified with a global coordinate system and variation of the heat source velocity direction is also introduced to consider the material separation point and the material above and below it.

3.2.1. Introduced global coordinate system

The global coordinate system is introduced by adding an offset on the heat sources as shown in Fig. 6. The offsets dx and dy impact the calculation of the point at l_i in the heat source (point $P(x_1, y_1)$ and its image $P'(x_2, y_2)$ in Fig. 6) along with the distances R and R' .

Evaluation of the distances R and R' with an adiabatic surface of equation $y = a_s x + b_s$ is given as:

$$x_1(l_i, \gamma, dx) = l_i \cos \gamma + dx \quad (21)$$

$$y_1(l_i, \gamma, dy) = l_i \sin \gamma + dy \quad (22)$$

$$R(x, y, l_i, \gamma, dx, dy) = \sqrt{(x - x_1(l_i, \gamma, dx))^2 + (y - y_1(l_i, \gamma, dy))^2} \quad (23)$$

$$x_2(l_i, \gamma, dx, dy) = \frac{2(x_1(l_i, \gamma, dx) + y_1(l_i, \gamma, dy)a_s - a_s b_s)}{1 + a_s^2} - x_1(l_i, \gamma, dx) \quad (24)$$

$$y_2(l_i, \gamma, dx, dy) = 2\left(a_s \frac{(x_1(l_i, \gamma, dx) + y_1(l_i, \gamma, dy)a_s - a_s b_s)}{1 + a_s^2} + b_s\right) - y_1(l_i, \gamma, dy) \quad (25)$$

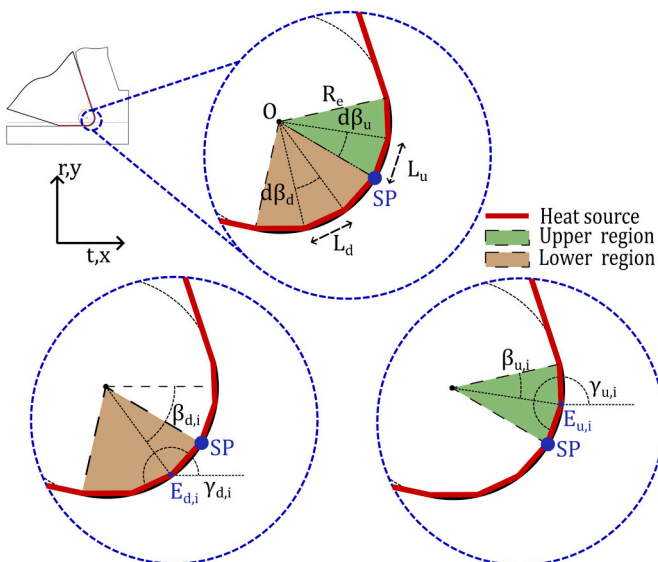


Fig. 5. Discretization of the edge radius

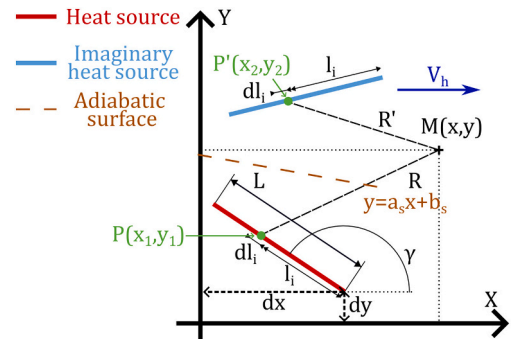


Fig. 6. Tilted and shifted moving line heat source.

$$R'(x, y, l_i, \gamma, dx, dy) = \sqrt{(x - x_2(l_i, \gamma, dx, dy))^2 + (y - y_2(l_i, \gamma, dx, dy))^2} \quad (26)$$

where l_i is the position of the differential heat element dl_i on the line heat source, γ is the slope of the line heat source, dx, dy are the offsets of the line heat source and a_s, b_s are the coefficients of the symmetry axis.

In case the symmetry axis is vertical, i.e. $x = a_s$, the following equations apply:

$$x_2(l_i, \gamma, dx, dy) = 2a_s - x_1(l_i, \gamma, dx) \quad (27)$$

$$y_2(l_i, \gamma, dx, dy) = y_1(l_i, \gamma, dy) \quad (28)$$

Similar equations can be obtained for the stationary rectangular heat sources by adding the third dimension (z) in the calculation of the distances R and R' as:

$$R(x, y, z, l_i, z_i, \gamma, dx, dy) = \sqrt{(x - x_1(l_i, \gamma, dx))^2 + (y - y_1(l_i, \gamma, dy))^2 + (z - z_i)^2} \quad (29)$$

$$R'(x, y, z, l_i, z_i, \gamma, dx, dy) = \sqrt{(x - x_2(l_i, \gamma, dx, dy))^2 + (y - y_2(l_i, \gamma, dx, dy))^2 + (z - z_i)^2} \quad (30)$$

3.2.2. Integration of the velocity direction

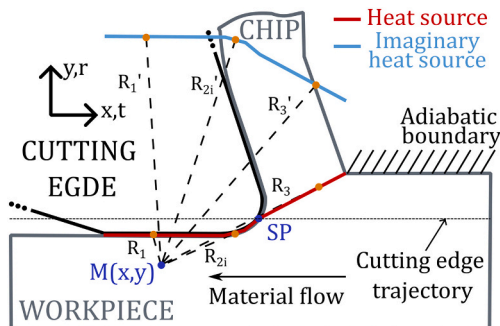
The velocity direction in existing studies cannot be changed because only the distance R between the point of interest $M(x, y, z)$ and the position at l_i in the heat source is projected in the direction of the heat source motion. It means that the contribution of the velocity on the moving imaginary heat source is not included in the temperature field when the heat source velocity direction is changed, which provides inconsistent results in such cases. This can be solved by separating the contribution of the original and imaginary heat sources as:

$$\theta(x, y) = \int_0^L B(l_i) \frac{q(l_i)}{2\pi\kappa} e^{-R_v V_h / 2aK_0} \left(\frac{V_h R}{2a} \right) dl_i + \int_0^L B(l_i) \frac{q(l_i)}{2\pi\kappa} e^{-R'_v V_h / 2aK_0} \left(\frac{V_h R'}{2a} \right) dl_i \quad (31)$$

where R_v and R'_v are respectively the projections of R and R' in the direction of motion.

3.3. Integration of flank wear

Temperature prediction equations are defined in the three following bodies of a worn tool: workpiece, chip and the cutting tool. The flank face of the tool is described by the flat wear length (VB). According to



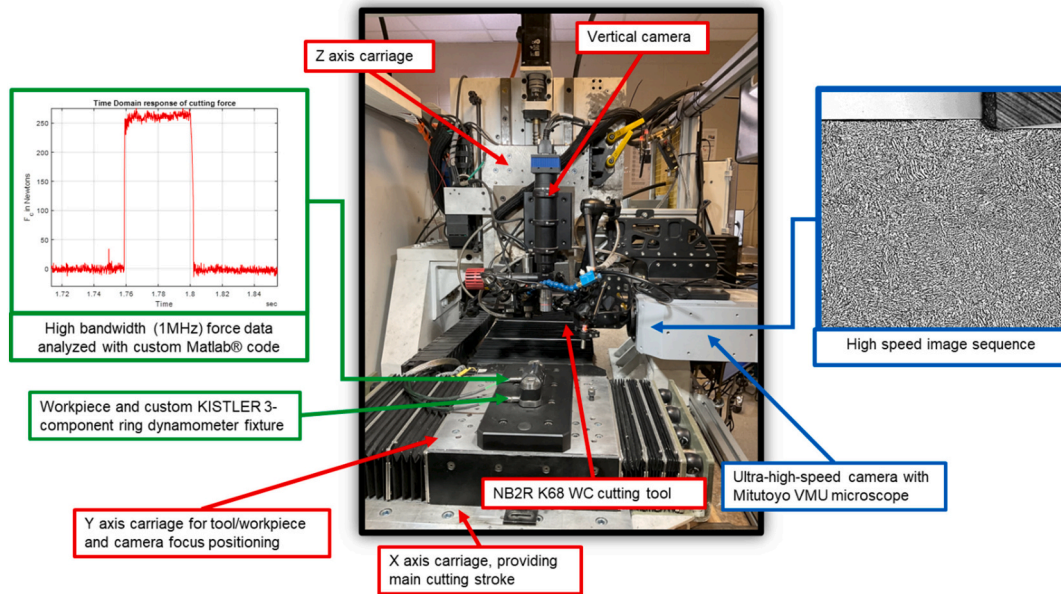


Fig. 10. Experimental setup for in-situ orthogonal cutting tests.

Table 4.1
Geometric parameters of the cutting edge.

Geometry	Rake angle, α_r [°]	Separation angle, χ [°] (arbitrary)	Wear length, VB [μm]	Edge radius, R_e [μm]
1	0	50	50	15
2			100	25
3			150	35
4			250	55

details on the technical specifications and operation of this setup are provided in [36,37].

4.1. Residual stress measurements and inverse method description

The peak workpiece temperature under the flank face, located at the extremity of the wear flat away from the edge radius, was inversely measured using the residual stress measurements for four different cutting edge and tool wear geometries (Table 4.1) and cutting parameters (Table 4.2). The uniform flank wear was artificially generated by grinding and diamond lapping (30 μm abrasive size) as a flat facet. The wear length was measured from the extreme tip of the cutting edge using a brightfield 3D microscope. The temperature-dependent material properties of Inconel 718 (workpiece) and the cemented carbide cutting tool are listed in Table 4.3.

The relationship between the peak surface tensile residual stress and the peak flank/workpiece temperature at the elastic recovery exit point was assumed to be directly related to the material specific response of

Table 4.3

Temperature dependent mechanical and thermal properties of Inconel 718 and the uncoated cemented carbide cutting tool.

Property	Inconel 718 (valid for $T \in [0, 1000]$ °C) [38]
Density: ρ (kg/m ³)	8220
Thermal conductivity: κ (W/(m·K))	$\kappa(T) = 0.01505 \cdot T + 11.07$
Specific heat: c_p (J/(kg·K))	$c_p(T) = 0.0002339 \cdot T^2 - 0.03916 \cdot T + 428$
Thermal diffusivity: a (m ² /s)	$a(T) = 7.292 \cdot 10^{-16} \cdot T^3 - 3.632 \cdot 10^{-12} \cdot T^2 + 4.296 \cdot 10^{-9} \cdot T + 3.134 \cdot 10^{-6}$
	Cemented carbide (valid for $T \in [0, 1200]$ °C) [39]
Thermal conductivity: κ (W/(m·K))	$\kappa(T) = -4.221 \cdot 10^{-6} \cdot T^2 + 0.017 \cdot T + 32.88$

Inconel 718 to localized thermal expansion and the associated deviatoric stress according to the relationship provided in [38] and shown in Eq. (38):

$$\sigma_t = \sigma_\alpha - \sigma_y = \frac{\Delta T \cdot E(T) \cdot \alpha_L(T)}{1 - \nu(T)} - \sigma_y \quad (38)$$

where $\sigma_\alpha(T)$ is the stress due to thermal expansion and $\sigma_y(T)$ represents the yield strength, both of which depend on temperature and the temperature differential between the peak and bulk temperature, ΔT . Likewise, the elastic modulus $E(T)$ and Poisson's ratio $\nu(T)$, as well as the coefficient of linear thermal expansion $\alpha_L(T)$ are functions of temperature, so the evolution of thermally-induced residual stress depends on several key material properties that are subject to change with

Table 4.2
Cutting parameters used in the machining tests.

Geometry	Uncut chip thickness, h [μm]	Cutting width, w [mm]	Cutting speed, V [m/min]	Chip thickness, h_c [μm]	Tangential cutting force, F_{tc} [N] (measured)	Radial cutting force, F_{fc} [N] (measured)	Radial ploughing force, F_p [N] (measured)
1	25	3	30	50	193	115.5	256.5
2					197	125.5	332.5
3					203.5	136	425
4					218	152	684

Where h_c is the cut chip thickness. A representative chip compression ratio (h/h_c) of 0.5 is assumed.

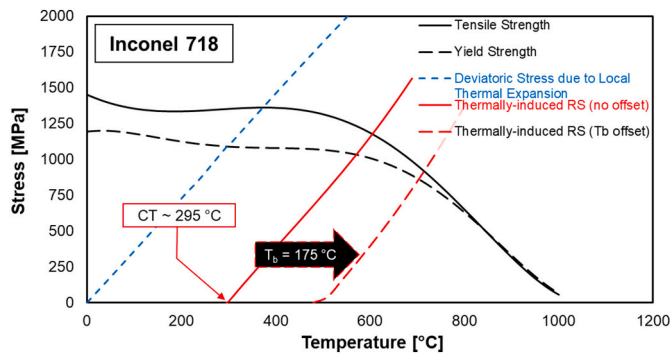


Fig. 11. Thermo-mechanical response of Inconel 718 to localized heating and resulting relationship between the peak surface temperature differential and tensile residual stress.

temperature. Fig. 11 illustrates how the increasing deviatoric stress due to localized temperature differences exceeds the yield strength of Inconel 718 at approximately 295 °C, beyond which a thermally-induced tensile residual stress (RS) will remain on the surface layer of the locally heated material.

As shown in Fig. 11, there exists nearly a linear (more accurately parabolic) relationship between the peak surface temperature and measured surface RS. It should be noted that the shown parabolic relationship assumes that the bulk temperature (T_b) of the Inconel 718 workpiece is zero degree Celsius, which is of course not the case in practice. To account for the bulk workpiece temperature, whether room temperature for a “cold” workpiece or some elevated temperature due to previous machining passes, the critical temperature (CT) and RS/temperature relationship can simply be offset by the bulk temperature value, as illustrated by the associated arrow and dashed red line in Fig. 11. For the present study, the bulk temperature for each feed/speed and wear condition was analytically determined as the temperature at 20 times the half width of contact from the tool flank exit point, rather than the average temperature of the entire workpiece. Fig. 12 illustrates the normalized tool flank/workpiece interface temperature profiles for three thermal (Péclet) numbers, ranging from steady-state (0.5), to mildly transient (5.0), and highly transient (50). As can be seen, the normalized temperature profiles approach an intermediate steady-state (i.e., quasi-bulk) temperature around 20 normalized half-widths from

the exit point ($x/b = 1$).

This assumption was adopted to reasonably represent the regional bulk temperature, rather than a global bulk temperature which has less physical relevance in the context of localized thermal expansion due to steep thermal gradients induced in the tool flank region. The result of this assumption was an additional offset between 150 and 200° Celsius for the various wear conditions, with higher wear conditions resulting in lower bulk temperature offset values due to the less pronounced thermal gradient and larger half width of the thermal contact. By plotting the inversely determined temperatures against the measured residual stress, the plot shown in Fig. 13 was generated. When considering the uncertainty of the proposed method, which is estimated to be on the order of $\pm 5\%$, the linear regression shown for the $VB = 150\ \mu\text{m}$ ($VB150$) condition can be reasonably applied for all worn tool conditions in the present study.

Surface stress measurements were obtained for each experimental condition using the X-ray diffraction (XRD) technique on a Rigaku Diffractometer with Cu K- α radiation (8.04 keV, $\lambda = 1.5406\ \text{\AA}$) via the $\sin^2\psi$ -method at the 220 peak (face-centered cubic, gamma prime phase at 75.2°). For analysis of the Inconel 718 workpiece material data, a lattice parameter of $3.607\ \text{\AA}$, Elastic modulus of 190 GPa, and Poisson's ratio of 0.294 were used. The penetration depth of the Cu K- α radiation at the 75.2° inclination angle, defined as the depth from which 90 % of the total measured signal intensity is obtained, was approximately $3\ \mu\text{m}$. The resulting stress measurements therefore represent a weighted average of the state of stress within this very shallow near-surface layer. As a result of this averaging, as well as due to experimental uncertainties and simplifying assumptions of the inverse stress/temperature correlation method, the overall uncertainty of the measured temperatures was estimated to be $\pm 12\%$, or approximately $\pm 50\ ^\circ\text{C}$ within the evaluated range.

The cutting forces in radial (F_f) and tangential (F_t) were measured for different uncut chip thickness (h) values and used to isolate the edge ploughing F_p and tangential edge F_{te} forces (Table 4.2) using the linear orthogonal machining mechanics [35] by extrapolating the measured forces to $h = 0\ \mu\text{m}$ as exemplified for $VB=50\ \mu\text{m}$, $R_e=15\ \mu\text{m}$ in Fig. 14. Though the edge forces are separated experimentally in this work (Fig. 14) one can use the slip-line theory based methods [40,41] to predict and use them in the proposed improved temperature prediction model.

Once the cutting and edge forces are obtained, a recursive method is

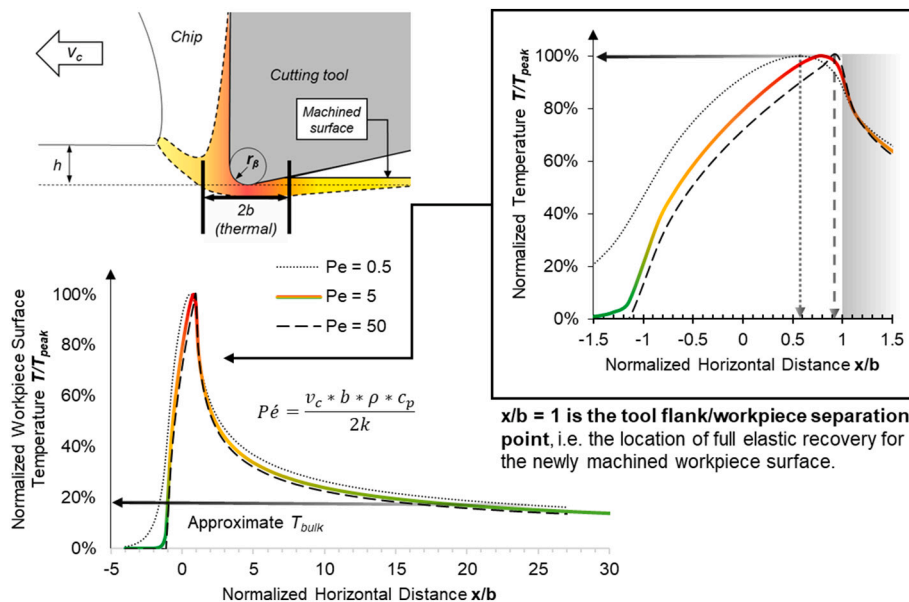


Fig. 12. Normalized tool flank/workpiece interface temperature profiles for three thermal (Péclet) numbers and illustration of quasi-bulk temperature location.

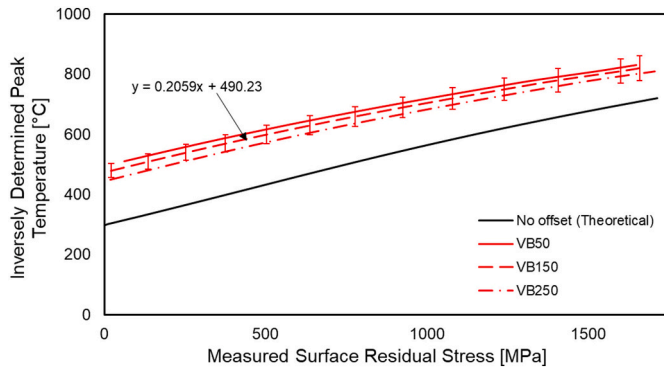


Fig. 13. Inverse relationship between the measured surface stress and peak tool flank/workpiece exit temperature with error bars indicating uncertainty of the proposed method.

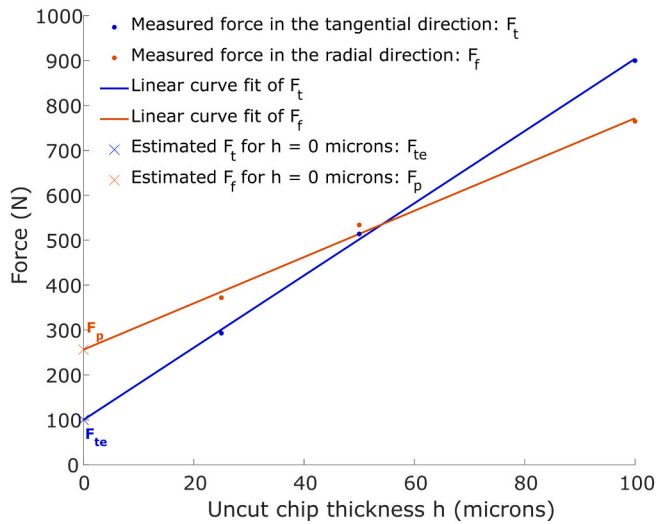


Fig. 14. Example of how edge ploughing F_p and tangential edge F_{te} forces are experimentally identified by extrapolating the total measured tangential and radial forces to $h = 0 \mu\text{m}$ (for $VB = 50 \mu\text{m}$, $R_e = 15 \mu\text{m}$).

adopted to calculate the temperature at the point of interest, using the temperature-dependent material properties of Inconel 718 listed in Table 4.3. This is achieved by assuming an initial temperature of 500°C (which has given the fastest convergence time), calculating the temperature at the point of interest, and repeating the process until a threshold of 10°C is reached. Convergence was systematically achieved after two iterations.

To avoid the inaccuracy in the workpiece temperature prediction described in Section 3.3.1, three linear heat sources are used to approximate the upper circular edge radius ($n_u = 3$) and six on the lower portion ($n_d = 6$). This is based on the convergence analyses performed by varying the number of heat sources on the upper and lower circular portions from 1 to 9 as shown in Figs. 15 and 16.

As shown in Fig. 15, the predicted temperature is sensitive to the number of heat sources in the lower region (n_d) with abnormal temperature values observed for $n_d = 1$. However, these discrepancies diminish significantly, and convergence is reached for $n_d \geq 3$, which led to a conservative choice of $n_d = 6$ in this work. In contrast, Fig. 16 indicates that the temperature at the point of interest is not sensitive to the value of n_u . That parameter would mostly affect the temperature on the rake face of the tool due to its localization and the material flow direction. A value of $n_u = 3$ is therefore chosen in this study.

It can also be noted that the separation angle (χ), which is considered constant at 50° in this work, varies in reality. A range of $[40^\circ, 50^\circ]$ can be

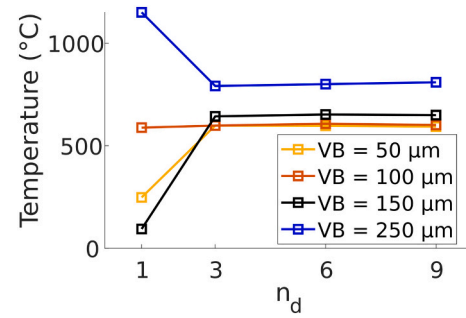


Fig. 15. Sensitivity analysis for the number of heat sources on the lower portion of the circular edge radius (n_d), with a constant $n_u = 3$, for all wear length (VB) values.

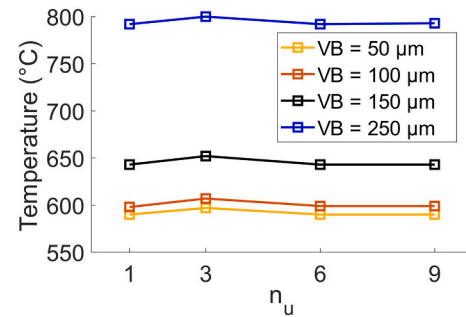


Fig. 16. Sensitivity analysis for the number of heat sources on the upper portion of the circular edge radius (n_u), with a constant $n_d = 6$, for all wear length (VB) values.

deduced from [25], and the model results have been calculated for three χ values: 40° , 45° and 50° , as presented in Fig. 17.

Fig. 17 shows that the separation angle (χ) do not affect the temperature results, except for $\chi = 40^\circ$ and $VB = 250 \mu\text{m}$. As the separation angle decreases, the lower edge portion becomes smaller, increasing the heat intensity in Eq. (5). That leads to inaccuracies in heat partition calculations due to imbalance between the small heat sources with large heat intensity and long heat sources with lower heat intensity. Therefore, an average separation angle of 50° is kept in model calculations.

Comparison of the experimentally measured peak workpiece temperatures and the analytically predicted ones using the proposed improved model is presented in Fig. 18.

As seen, the predictions of the proposed improved analytical model are in good agreement with the experimentally measured peak workpiece temperatures with an average discrepancy of 18 % and a maximum discrepancy of 24 % for the extreme wear length of $VB = 250 \mu\text{m}$. Some constant offset between the measurements and theory is to be expected, since Kommanduri's energy conversion factor of 87.5 % is adopted in this study. The increasing prediction error as the wear length (VB) and edge radius (R_e) increase can be attributed to two factors. First, the linear approximation of the edge radius may lead to inaccurate results when the edge radius is large (in our case, $R_e = 55 \mu\text{m}$) due to the formation of dead metal zone that is not included in the proposed model. Secondly, increase in the wear length leads to a larger contact surface, which directly influences heat partitioning. More precisely, the presence of multiple small heat sources at the edge radius combined with a single long heat source at the tool's flank cause inaccuracies in the heat partition calculations. The small but intense heat sources in the flat wear interface disproportionately affects the heat distribution. This last point is also the reason for the constant overestimation of the temperature. Overall, the prediction accuracy is satisfactory and close to the uncertainty range of the experimental measurements for the given wear length range of $VB \in [50, 250]$.

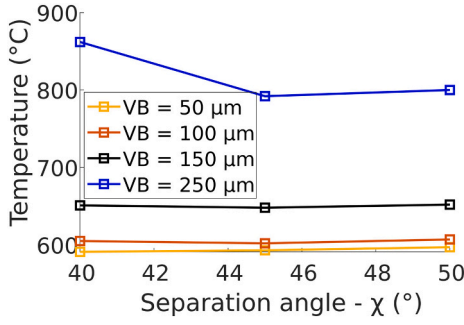


Fig. 17. Sensitivity analysis for the separation angle (χ) with $n_d = 6$ and $n_u = 3$, for all wear length (VB) values.

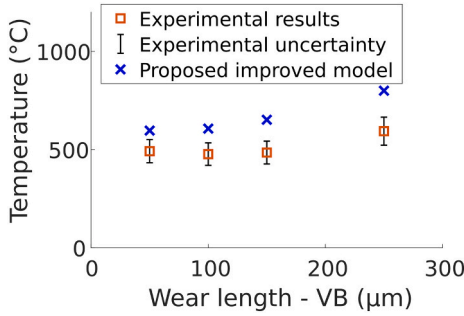


Fig. 18. Peak workpiece temperature: comparison of the experimental and analytically predicted results.

Furthermore, each simulation case took under 5 min on a computer with an AMD Ryzen 7 3700X processor and 8 cores when the edge radius and wear length are both considered. Removing the effect of edge radius reduces the computation time to 17 s, and further removing the flat wear reduces it to 8 s on the same computer. It is evident that the improved analytical model is a computationally efficient alternative to costly and time-prohibitive experimental or FE-based methods, which require complex data processing.

5. Conclusion

This article proposes an improved analytical temperature prediction model for orthogonal machining by considering the effect of edge radius and the flank wear length, and the coupling between two (edge radius changes as a function of tool wear). The main contribution is the integration of the honed edge radius R_e in the temperature prediction for the first time in literature. This was motivated by its high value (R_e up to 55 μm) in the experiments conducted within this work. The inclusion of the hone radius required the introduction of a general coordinate system and the ability to vary the direction of the heat source velocity, which are the principal additions of the improved model. This enables more flexible and less complex mathematical formulations for the temperature calculations. The modified temperature model has been experimentally verified, and its predictions match well with the experimental results. However, decrease of the wear length under 50 μm may increase the temperature prediction error. This is because the heat partition calculation is unbalanced in case of a small heat source with dominant heat generation. Future work is needed to explore the limitations of the model in these specific cases more thoroughly, but the model can still be used as an efficient alternative to the experimental temperature measurements or numerical simulations for orthogonal machining. Replacing the linear approximation of the edge radius with a circular heat source, coupling the cutting force prediction model with the temperature prediction model to eliminate the need for experimental forces, and

expanding the model to oblique cutting are within the scope of future research.

Nomenclature

a	Thermal diffusivity
$A_{IJ}(i, j)$	Temperature rise at l_i at the interface I due to the heat source element $[l_j, l_{j+1}]$ in the heat source J and its corresponding equations
a_s, b_s	Parameters of the linear equation for a symmetry axis
$B(l_i)$	Heat partition of a heat source at l_i
$B_I(i)$	Heat partition at l_i in the interface I of two corresponding heat sources
$B_m, (1 - B_m)$	Heat partition of two heat source elements $[l_j, l_{j+1}]$ on both sides of the interface m
$C_I(i)$	Temperature rise at l_i at the interface I of all heat sources being the constant terms of the linear equation
$c_p(T)$	Temperature dependent specific heat
dx, dy	Offset of line heat source in the tangential and radial directions
$d\beta_q$	Angle between two successive tilted line heat sources approximating a circle
E	Modulus of elasticity
F_p	Ploughing force in the radial direction
F_{tc}, F_{fc}	Tangential and feed cutting forces
h	Uncut chip thickness
h_c	Cut chip thickness
$K_0(x)$	Bessel function of second kind of order 0 evaluated at x
L	Heat source length
l_i	Position in a heat source
L_m	Length of the heat source m
n	Number of discretized heat source elements
n_d, n_u	Number of linear heat sources approximating the lower and upper regions of the edge radius
P_x, P_y	x and y -coordinates of a point P
$q(l_i)$	Heat intensity of the heat source at l_i
$q_{f,r}(l_i)$	Non-uniform heat intensity of the rake-chip interface
q_r	Uniform heat intensity of the rake-chip interface
r, t, a	Radial, tangential and axial directions of the cut
R, R'	Distances between a point of interest $M(x, y, z)$ and the position at l_i in the original and imaginary heat sources
R_v, R'_v	Projection of R and R' in the direction of motion
R_e	Edge (hone) radius
r_s	Constant defining the stick and slide portion on the rake-chip interface
T	Temperature of the work material and in the tool
T_{init}	Initial temperature of the work material and the tool
V	Cutting speed
VB	Flat flank wear length
V_h	Heat source velocity
w	Width of cut
x_1, y_1	Coordinate of the point at l_i in a heat source
x_2, y_2	Coordinate of the point at l_i in an imaginary heat source
α_r	Rake angle
α_L	Coefficient of linear thermal expansion
β_a	Average friction angle
$\beta_{d,i}$	Angle of line of the i^{th} tilted heat source approximating a circle, with respect to the tangent direction
γ_m	Angle of the tilted heat source m
$\sigma_a(T)$	Stress due to thermal expansion
$\sigma_y(T)$	Yield strength
$\theta(x, y, z)$	Temperature rise at a point P defined in the Cartesian coordinates x, y, z
$\theta_{bm}(l_i)$	Temperature rise at l_i on the side b from the heat source at the interface m

$\kappa(T)$	Temperature dependent thermal conductivity
ρ	Density
ν	Poisson's ratio
ϕ	Shear angle
χ	Separation angle

CRedit authorship contribution statement

Jonathan Theraroz: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Oguzhan Tuysuz:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Julius Schoop:** Writing – review & editing, Resources, Methodology, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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